

XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Week 15: Revision

By:
Dr. Ng Zi Neng
Email: zineng.ng@xmu.edu.my

Scope of Final Exam

- Total of 6 questions
  - Q1 (20 marks): 5 conceptual questions on **random** topics, explain a situation using relevant physical theories, or derive an equation.
  - Q2 (15 marks): Calculation/Problem solving questions from **Week 1** to **Week 5** topics.
  - Q3 (20 marks): Calculation/Problem solving questions from **Week 7** topics.
  - Q4 (15 marks): Calculation/Problem solving questions from **Week 8** and **Week 9** topics.
  - Q5 and Q6: Calculation/Problem solving questions from **Week 10** and **Week 13** topics.

Revision questions

1. Conceptual problems:
   a. In uniform circular motion, how does the acceleration change when the speed is increased by a factor of 3? When the radius is decreased by a factor of 2?
   b. Can a body be in equilibrium when only one force acts on it? Explain.
   c. Why are cars designed to crumple in front and back for safety? Explain.
   d. Is it possible for a friction force to increase the mechanical energy of a system? Explain.
   e. The harder you hit the brakes while driving forward, the more the front end of your car will move down (and the rear end move up). Why?
   f. The proportions of various gases in the earth's atmosphere change somewhat with altitude. Would you expect the proportion of oxygen at high altitude to be greater or less than at sea level compared to the proportion of nitrogen? Why?

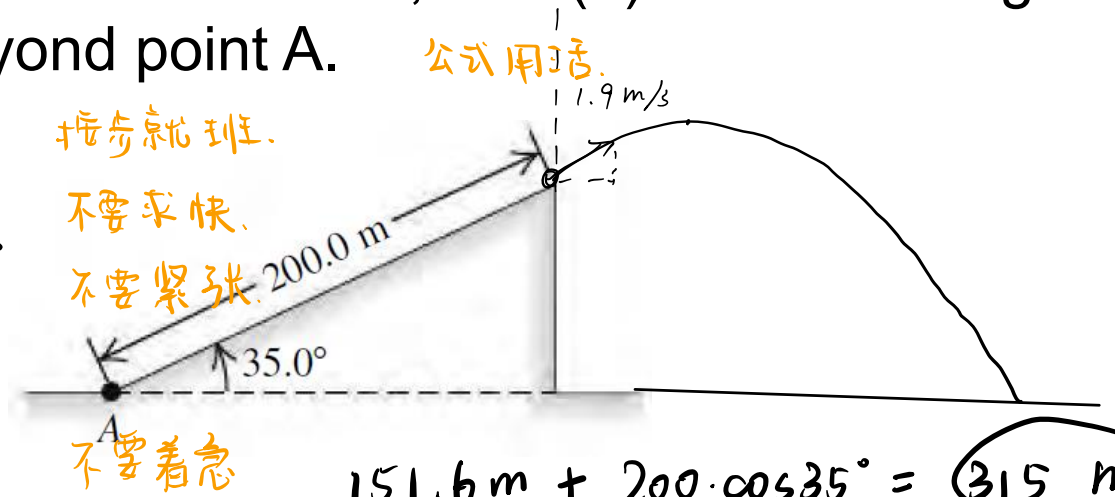
Revision questions

搞清楚手上的数据是什么.

2. A test rocket starting from rest at point A is launched by accelerating it along a 200.0-m incline at 1.90 m/s^2. The incline rises at 35.0° above the horizontal, and at the instant the rocket leaves it, the engines turn off and the rocket is subject to gravity only (ignore air resistance). Find (a) the maximum height above the ground that the rocket reaches, and (b) the rocket's greatest horizontal range beyond point A.

128 m.

公式用活.



按步就班.

不要求快.

不要紧张.

不要着急

$1.9\,m/s \cdot \sin 35° = V_y = 1.09\ m/s$

$V_y = g t_1 \Rightarrow t_1 = \frac{V_y}{g} = 0.11\ s$

$S_1 = \frac{1}{2} g t_1^2 = 0.059\ m$

$x_1 = 200 \cdot \sin 35° + S_1 = 114.77\ m$

$x = \frac{1}{2} g t_2^2 \Rightarrow t_2 = 4.84\ s$

$151.6\ m + 200 \cdot \cos 35° = 315\ m$

$v_x = 1.9 \cdot \cos 35° = 1.56$ m/s

$x = v_x \cdot (t_1 + t_2) = 7.7$

$x_2 = 200 \cdot \cos 35° + x = 171.535$ m.

Revision questions

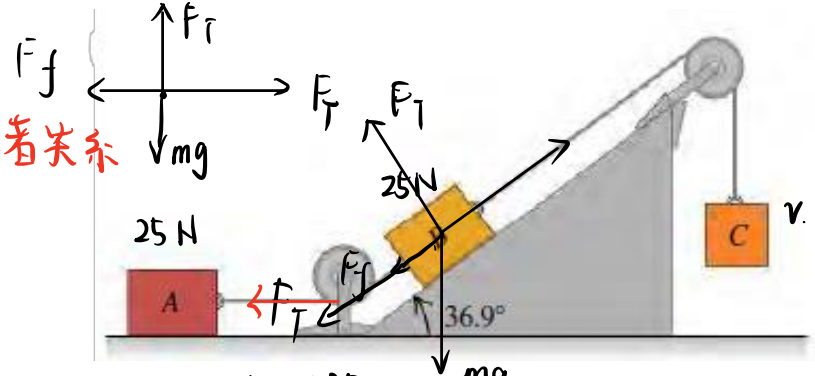
3. Blocks A, B, and C are placed as in figure and connected by ropes of negligible mass. Both A and B weigh 25.0 N each, and the coefficient of kinetic friction between each block and the surface is 0.35. Block C descends with constant velocity. (a) Draw separate free-body diagrams showing the forces acting on A and on B. (b) Find the tension in the rope connecting blocks A and B. (c) What is the weight of block C? (d) If the rope connecting A and B were cut, what would be the acceleration of C?

认真分析，一步错，步步错…一定想清楚2者关系

$F_{f_1} = \mu \cdot F_T = 0.35 \times 25 = 8.75$ N

$= F_{T_1}$

$mg\sin 36.9° + \mu mg\cos 36.9° + 8.75\text{N} = G_C$



$= 30.76\,N$ ✓

$\mu = 0.35$

$(m_b + m_c)a \Rightarrow a = ?$

Revision questions

$F_{f_1} = 8.75\,N = m_c a \Rightarrow a = 2.788\,m/s^2$ 1.54 m/s²

1.2kg.

4. (Past year question) A spring of negligible mass has force constant k = 1600 N/m. (a) Solve for the distance of the compression for 3.20 J of potential energy to be stored in the spring. (b) The uncompressed spring is placed vertically with one end on the floor, a 1.20-kg book is then dropped onto it from a height of 0.800 m above the top of the spring. Compute the maximum distance the spring will be compressed, illustrate this system with a simple diagram.

0.8m.

a) $E = \frac{1}{2}kx^2 = \frac{1}{2} \cdot 1600 \cdot x^2 = 3.2$

$x = 0.063\,m$

不要经验主义

$\sqrt{0.004} \neq 0.2$

$\frac{1}{2} \times 1600x^2 - 1.2 \times 9.8 \times x - 1.2 \times 9.8 \times 0.8 = 0$

$800x^2 - 11.76x - 9.408 = 0$

$\Rightarrow x =$

0.116 m.

b) $mgh = \frac{1}{2}kx^2 = 1.2 \times 9.8 \times 0.8$

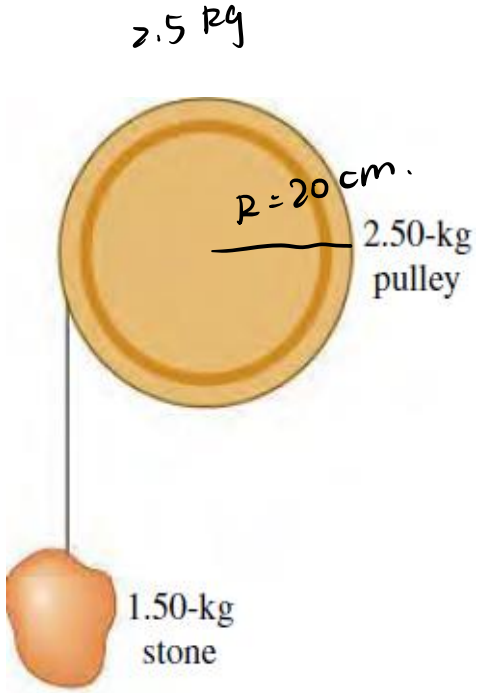
$\Rightarrow x = 0.108\,m$

$mg(h+x) = \frac{1}{2}k \cdot x^2$

不要觉得自己很对.实际上有很多应该注意的点.

Revision questions

5. A frictionless pulley has the shape of a uniform solid disk of mass 2.50 kg and radius 20.0 cm. A 1.50-kg stone is attached to a very light wire that is wrapped around the rim of the pulley, and the system is released from rest. (a) How far must the stone fall so that the pulley has 4.50 J of kinetic energy? (b) What percent of the total kinetic energy does the pulley have?



a) $K = 4.5\,J = \frac{1}{2} I \omega^2$.

$\Rightarrow \omega =$

$\Rightarrow v = \omega \cdot r$

$K + \frac{1}{2} m v^2 = mgh$

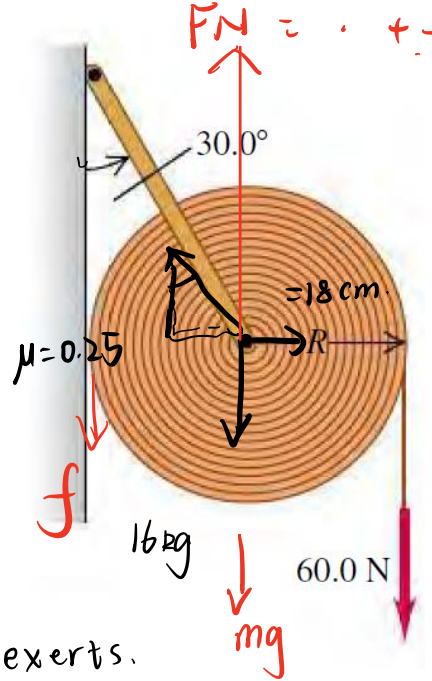
(b) $\frac{K}{mgh} \times 100\% =$ 45.5%

⇒ h = 0.673 m.

Revision questions

$F_N \cdot \sin 30° = f$.

6. A large 16.0-kg roll of paper with radius R = 18.0 cm rests against the wall and is held in place by a bracket attached to a rod through the center of the roll. The rod turns without friction in the bracket, and the moment of inertia of the paper and rod about the axis is 0.260 kg.m^2. The other end of the bracket is attached by a frictionless hinge to the wall such that the bracket makes an angle of 30.0° with the wall. The weight of the bracket is negligible. The coefficient of kinetic friction between the paper and the wall is μk = 0.25. A constant vertical force F = 60.0 N is applied to the paper, and the paper unrolls. What is the magnitude of (a) the force that the rod exerts on the paper as it unrolls; (b) the angular acceleration of the roll?



exerts.

$\alpha = 0. = \tau \cdot r$

$F_{rod} = 293$ N

$16 \times 9.8 \times \tan 30° = 90.529$ N > 60 N

Revision questions

7. A copper calorimeter can with mass 0.446 kg contains 0.0950 kg of ice. The system is initially at 0.0°C. (a) If 0.0350 kg of steam at 100.0°C and 1.00 atm pressure is added to the can, what is the final temperature of the calorimeter can and its contents? (b) At the final temperature, how many kilograms are there of ice, how many of liquid water, and how many of steam?

a) 0.035 kg 100.0°C 1atm. 要考虑 copper calorimeter.

$Q = m \cdot L_f$

$m_{water} = 9.13$ kg

Revision questions

8. A sealed box contains a monatomic ideal gas. The number of gas atoms per unit volume is 5.00×10^{20} atoms/cm^3, and the average translational kinetic energy of each atom is 1.80×10^{-23} J. (a) What is the gas pressure? (b) If the gas is neon (molar mass 20.18 g/mol), what is v_{rms} for the gas atoms?

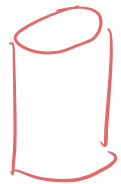
P

6×10^3 Pa.

32.8 m/s.

Revision questions

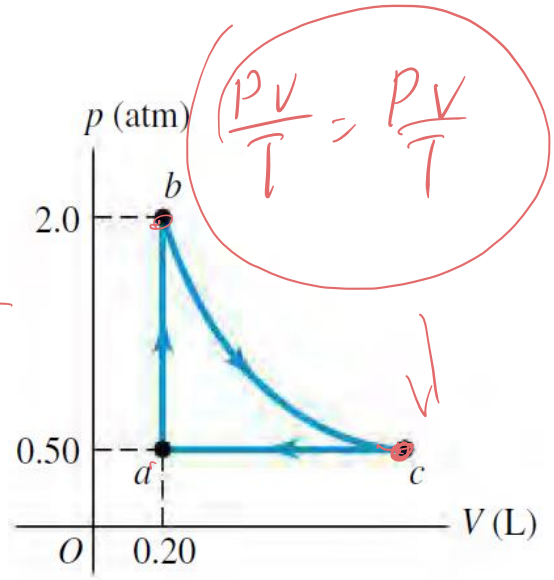
9. (Past year question) A cylinder 1.00 m tall with inside diameter 0.120 m is used to hold propane gas (molar mass 44.1 g/mol) for use in a barbecue. It is initially filled with gas until the gauge pressure is 1.30 × 106 Pa at 22.0°C. The temperature of the gas remains constant as it is partially emptied out of the tank, until the gauge pressure is 3.40 × 105 Pa. Calculate the mass of propane that has been used.



0.195 kg.

Revision questions

10. Figure shows a pV-diagram for 0.0040 mol of ideal H_2 gas. The temperature of the gas does not change during segment bc. (a) What volume does this gas occupy at point c? (b) Find the temperature of the gas at points a, b, and c. (c) How much heat went into or out of the gas during segments ab, ca, and bc? Indicate whether the heat has gone into or out of the gas. (d) Find the change in the internal energy of this hydrogen during segments ab, bc, and ca. Indicate whether the internal energy increased or decreased during each segment.



$\frac{PV}{T} = \frac{PV}{T}$

1. 0.8 L

2. 325 K

1220 K 1220 K

3. 76J

-107J

56J

4. +76J.